

C) Authors achieve hybridization by physically connecting piezoelectric cavity and silicon OMC cavity using a specialized phonon waveguide segment

This waveguide is created by gradually altering the dimensions of the holes in the section of the silicon nanobeam bridging two cavities. Thus, acoustic band structure is shifted to close the photonic bandgap at 56 THz allowing modes to freely propagate between two cavities. These hybridized modes have displacement profiles that span the entire structure, meaning a phonon generated at IDT travels through the waveguide and populates the OMC defect where it can interact with the optical field.

→ See page 2 second column

This physically separated intermediate structure is necessary because high energy telecom laser light used in OMC would immediately break Cooper pairs in a directly adjacent superconducting transmon. This would rapidly destroy the qubits fragile coherence. Furthermore, high-efficiency optomechanics requires material with high refractive index (like Si) but Si is non-piezoelectric. Conversely, strong electrical coupling requires a piezoelectric material (like AlN) which may not possess ideal refractive properties for a high-Q optical cavity. The modular design allows authors to indep. optimize piezoelectric transducer and optomech. cavity without compromising either's performance

→ See page 1, second column and page 2 second column

Workshop Problem 1

- A) A) OMC crystals realize co-localized modes by using a periodic silicon nanobeam patterned with specifically dimensioned holes that act as a simultaneous mirror for both light (~ 200 THz) and sound (5 GHz). By introducing a defect region where the hole dimensions and spacing are gradually tapered, the local band structure is altered to trap both photons and phonons in the same microscopic volume.

Colocalization is essential for achieving a large optomechanical coupling rate (g_0). $g_0 = (\partial \omega_c / \partial x) X_{\text{ZPF}}$ dictates how much cavity freq. shifts due to a single phonon. By co-localizing modes in a sub-wavelength volume, effective mass of mechanical mode is minimized, (which increases X_{ZPF}), and optical force per unit displacement is maximized. This allows system to reach high-cooperativity regime necessary for coherent state transfer
→ See extended Data Fig. 2 and Fig. 3

- B) The transmon qubit couples to the piezoelectric resonator through a resonant interaction driven by piezoelectric effect. Transmons large capacitive leads are wired to an interdigital transducer (IDT) which consist of periodic aluminium electrodes patterned on a thin film of piezoelectric AlN (page 2, second column). When transmon is in excited state, the resulting microwave freq. oscillating E-field is applied across the IDT. E-field is then transformed into a periodic mechanical strain, launching a Lamb wave. By spatially matching the period of IDT fingers to the acoustic wavelength, electrical excitation is resonantly converted into mechanical phonons.

→ See back of the page

Workshop Problem 2: Optomechanical Interaction

A) Generalized interaction Hamiltonian is $H_{int} = \hbar G (\hat{b} + \hat{b}^\dagger) (\hat{a}^\dagger \hat{a})$. However when we apply the rotating wave approximation ($\hat{a} \rightarrow \alpha + \hat{a}$), under these 2 conditions:

1) Red-detuned driving: System is driven by a pump laser tuned to red-side band $\Rightarrow \omega_L = \omega_c - \omega_m$
 $\hookrightarrow$ See fig. 2 caption

2) Resolved-side band regime: Mechanical freq. must be much larger than optical cavity linewidth $\Rightarrow \omega_m \gg \kappa_0$

$$\omega_m / 2\pi = 5.16 \text{ GHz} \gg \kappa_0 / 2\pi = \kappa_{i,0} / 2\pi + \kappa_{e,0} / 2\pi = 0.8 + 0.81 = 1.61 \text{ GHz}$$

$\downarrow$ Fig. 2 caption / page 4, 1st column $\downarrow$ page 3, column 2

B) Optical Mode: Dissipation is dominated by intrinsic cavity losses and coupling to waveguide. Their rates are nearly identical
 $\hookrightarrow \kappa_{i,0} / 2\pi = 0.8 \text{ GHz}$ and $\kappa_{e,0} / 2\pi = 0.81 \text{ GHz}$ (pg. 3, 2nd column)

Mechanical Mode: Dissipation is strictly dominated by mechanical losses, $\Rightarrow$ energy decay $\kappa_{m,T1} / 2\pi = 446 \text{ kHz}$ and spectral diffusion contributing to a total linewidth of $\kappa_{i,m} / 2\pi = 1 \text{ MHz}$. (pg. 4, 1st column)
These intrinsic losses are much larger than optically induced back-action damping rate $\Rightarrow \gamma_{om} / 2\pi = 19 \text{ kHz}$ (pg. 3, 2nd column)

C) Authors use a pulsed scheme in contrast to CW operation studied in the lecture. Theoretical conversion eff. is:

$$\eta_k = \frac{\chi_{e,0}}{\chi_{i,0} + \chi_{e,0}} = \frac{0.81}{0.8 + 0.81} \approx 0.503 \quad : \text{ extraction efficiency} \quad \left(\text{formula in pg. 7, 2nd column} \right)$$

where $\chi_{i,0} = 0.80 \text{ GHz} \cdot 2\pi$: intrinsic cavity loss
 $\chi_{e,0} = 0.81 \text{ GHz} \cdot 2\pi$: coupling to waveguide } pg. 3 2nd column

$$\eta_{ro}(\tau_0) = \frac{\gamma_{om}}{\gamma_{om} + \chi_{m,\tau_1}} \left[1 - e^{-(\gamma_{om} + \chi_{m,\tau_1})\tau_0} \right] : \text{ internal readout efficiency} \quad \left(\text{formula in pg 7, 2nd column} \right)$$

$$= \frac{19}{465} \left(1 - e^{-(2.96 \times 10^6)(38 \times 10^{-9})} \right) \approx 4.28 \times 10^{-3}$$

where $\gamma_{om}/2\pi = 19 \text{ kHz}$ (pg 7), $\chi_{m,\tau_1}/2\pi = 446 \text{ kHz}$ (pg 3), $\tau_0 = 38 \text{ ns}$ (pg 4)

$$\eta_{ro\text{-total}} = \eta_k \eta_{ro}(\tau_0) \underbrace{\eta_{sys}}_{\approx 1.5 \text{ (pg 7)}} = 4.28 \times 10^{-3} \times 0.503 \times 1.5 \times 10^{-2} \approx \boxed{3.2 \times 10^{-5}}$$

→ Reported total readout eff. is $\eta_{ro\text{-reported}} = (1.18 \pm 0.26) \times 10^{-5}$

↳ Discrepancy might result from finite rise/fall time of square pulse

D) To increase conversion efficiency, system must be pushed deeper into the high cooperativity regime ($\gamma_{om} \gg \chi_{m,\tau_1}$)

1) Reduce intrinsic mechanical loss (χ_{m,τ_1}): Mitigate spectral diffusion and surface defects to lower mech. damping rate

2) Increase optomech. coupling (γ_{om}): Utilize material with higher inherent piezoelectric coupling rates (LiNbO_3)

3) Allow higher power pump (η_c): By improving surface passivation and better thermal dissipation of acoustic mode to cold environment, we can utilize a stronger pump drive without adding thermal noise

Workshop Problem 3: The Qubit-Mechanics Swap Operation

$$A) \quad \hat{H}_{pe}/\hbar = g_{pe} \left(\underset{\substack{\downarrow \\ \text{coupling} \\ \text{coeff.}}}{\hat{\sigma}_{eg}} \underset{\substack{\downarrow \\ \text{raising op.}}}{\hat{b}_m} + \hat{\sigma}_{ge} \underset{\substack{\downarrow \\ \text{lowering op.}}}{\hat{b}_m^\dagger} \right)$$

→ At the start of swap sequence ($t=0$), the qubit is prepared in an excited state $|e\rangle$ and mech. resonator in ground state $|0\rangle$

$$\hookrightarrow |\Psi(0)\rangle = |e, 0\rangle$$

$$\hat{H}_{pe} = \hbar \begin{pmatrix} 0 & g_{pe} \\ g_{pe} & 0 \end{pmatrix} \text{ in a subspace of } \{|e, 0\rangle, |g, 1\rangle\}$$

$$|\Psi(t)\rangle = e^{-i\hat{H}/\hbar} |\Psi(0)\rangle = \cos(g_{pe}t) |e, 0\rangle - i \sin(g_{pe}t) |g, 1\rangle$$

Prob. of finding the system in state $|g, 1\rangle$ is:

$$P_m(t) = |\langle g, 1 | \Psi(t) \rangle|^2 = \sin^2(g_{pe}t) \Rightarrow \text{Vacuum Rabi oscillation}$$

For optimal swap, we need to maximize $P_m(T_{\text{swap}})$

$$\hookrightarrow g_{pe} T_{\text{swap}} = \frac{\pi}{2} \Rightarrow g_{pe}/2\pi = \frac{1}{4 T_{\text{swap}}}$$

$$\text{With } T_{\text{swap}} = 104 \text{ ns (reported)} \Rightarrow \boxed{g_{pe}/2\pi \approx 2.4038 \times 10^6 \text{ Hz}}$$

B) Coherent swap operation is not instantaneous, It requires 104 ns to fully move the excitation from qubit to mechanical oscillator. During this 104 ns's, excitation exist as a superposition shared btwn 2 modes. Simultaneously, the mechanical mode is continuously dissipating energy to the environment, characterized by phonon lifetime of $T_1 = 357 \text{ ns} \Rightarrow$ Irreversible energy loss during state transfer.

$$\hookrightarrow e^{-104/357} \approx 0.747 \text{ almost perfectly matches with observed success rate of } 75\%$$

Problem 4: Added Noise and Bandwidth

A) Eq. 15 describes added noise for 2-stage transducer. Since this exp. utilizes a single optomech. readout port, we drop the 2nd optical port ($\gamma_{om,2}$)

$$n_{\text{added}} = \frac{\gamma_i n_b + \gamma_{om} \left(\frac{\chi}{4\omega_m} \right)^2}{h\chi \gamma_{om}}$$

→ At the dilution fridge's base temp. ($T \approx 15 \text{ mK}$), thermal phonon occupancy:

$$n_b = \frac{1}{e^{\frac{hf}{k_B T}} - 1} \approx 7 \times 10^{-8} \text{ phonons at } f = 5.16 \text{ THz}$$

↳ This noise is so small that, added noise should theoretically be dominated by quantum backaction from unresolved sidebands.

$$n_{\text{added}} \approx \frac{1}{h\chi} \left(\frac{\chi}{4\omega_m} \right)^2 \approx \frac{1}{0.503} \left(\frac{1.61 \text{ GHz}}{4 \times 5.159 \text{ GHz}} \right)^2 \approx 0.012 \text{ photons}$$

where $\chi/2\pi = 1.61 \text{ GHz}$, $h\chi = \frac{0.81}{1.61} \approx 0.503$, $\omega_m = 5.159 \text{ GHz}$

→ Calculated noise is drastically lower than experimental reported value (pg 7) of $n_{\text{added}} = 0.57 \pm 0.2$. As authors discuss, this discrepancy occurs because the system is not in thermal equilibrium during readout. Dominant source of measured noise is optical absorption heating caused by laser pulse which effectively raises the local bath occupancy far above base temperature.

B) → To increase efficiency (η_{rp}), one must increase the optical pump power (to increase γ_{om}) or extend duration of readout pulse. However, since silicon absorbs a fraction of this telecom laser light, pumping system harder increases local heating. This results in an increased thermal phonon occupancy which destroys the fidelity of fragile quantum state being transferred. Authors deliberately limit their readout to a short 38 ns windows to keep the added noise near the threshold, sacrificing efficiency in the process.

c) $BW = \gamma_i + \gamma_{om} \Rightarrow$ determined by intrinsic and mechanical linewidth (eq. 14, lecture 13)

$\rightarrow \gamma_i = \kappa_{i,m} / 2\pi = 1 \text{ MHz}$: intrinsic mech linewidth (pg 4, 2nd column)

$\rightarrow \gamma_{om} / 2\pi = 19 \text{ kHz}$: optomech. damping rate (pg 7, 2nd column)

$$BW = 1.019 \text{ MHz}$$

$$1) \quad \frac{dp}{dt} = i\varepsilon [a + a^\dagger, p] + \frac{\gamma}{2} (2ap a^\dagger - a^\dagger a p - p a^\dagger a)$$

$$\frac{d\langle a \rangle}{dt} = \text{tr} \left(a \frac{dp}{dt} \right) = i\varepsilon \text{tr} (a [a + a^\dagger, p]) + \frac{\gamma}{2} \text{tr} (2a^2 p a^\dagger - a a^\dagger a p - a p a^\dagger a)$$

$$\hookrightarrow \text{tr} (a, [a, p]) = \text{tr} (a^2 p) - \text{tr} (a p a) = 0 \quad (\text{Using } \text{tr}(AB) = \text{tr}(BA))$$

$$\hookrightarrow \text{tr} (a, [a^\dagger, p]) = \text{tr} (a a^\dagger p - a p a^\dagger) = \text{tr} ([a, a^\dagger] p) = \text{tr} (p) = 1$$

$$\hookrightarrow \frac{\gamma}{2} \text{tr} (2a^\dagger a^2 - a a^\dagger a - a^\dagger a^2) p = \frac{\gamma}{2} \text{tr} ([a^\dagger, a] a p) = -\frac{\gamma}{2} \text{tr} (a p) = -\frac{\gamma}{2} \langle a \rangle$$

$\Downarrow$
 cyclic permutations applied

$$\frac{d\langle a \rangle}{dt} = i\varepsilon - \frac{\gamma}{2} \langle a \rangle \Rightarrow \quad \text{In steady state } \frac{d\langle a \rangle}{dt} = 0 \text{ so,}$$

$$\langle a \rangle_{ss} = \frac{2i\varepsilon}{\gamma} \quad \checkmark$$

Now let us calculate $\frac{d\langle a^\dagger a \rangle}{dt} = \text{tr} \left(a^\dagger a \frac{dp}{dt} \right)$

$$\frac{d\langle a^\dagger a \rangle}{dt} = i\varepsilon \text{tr} (a^\dagger a [a + a^\dagger, p]) + \frac{\gamma}{2} \text{tr} (2a^\dagger a^2 p a^\dagger - a^\dagger a a^\dagger a p - a^\dagger a p a^\dagger a)$$

$$\hookrightarrow \text{tr} (a^\dagger a [a, p]) = \text{tr} (a^\dagger a^2 p - a^\dagger a p a) = \text{tr} ([a^\dagger, a] a p) = -\langle a \rangle$$

$$\hookrightarrow \text{tr} (a^\dagger a [a^\dagger, p]) = \text{tr} (a^\dagger a a^\dagger p - a^\dagger a p a^\dagger) = \text{tr} (a^\dagger [a, a^\dagger] p) = \langle a^\dagger \rangle$$

$$\hookrightarrow \text{tr} (2a^\dagger a^2 p a^\dagger - a^\dagger a a^\dagger a p - a^\dagger a p a^\dagger a)$$

$$= \text{tr} (2(a^\dagger^2 a^2 - a^\dagger a a^\dagger a) p) = \text{tr} (2(a^\dagger^2 a^2 - a^\dagger^2 a^2 - a^\dagger a) p)$$

$$= -2 \text{tr} (a^\dagger a p) = -2 \langle a^\dagger a \rangle$$

$$\frac{d\langle a^\dagger a \rangle}{dt} = i\varepsilon (-\langle a \rangle + \langle a^\dagger \rangle) - \gamma \langle a^\dagger a \rangle$$

$$\hookrightarrow \text{In steady state } \frac{d\langle a^\dagger a \rangle}{dt} = 0 \Rightarrow \langle a^\dagger a \rangle_{ss} = \frac{i\varepsilon}{\gamma} (\langle a^\dagger \rangle_{ss} - \langle a \rangle_{ss})$$

$$\text{Recall } \langle a \rangle_{ss} = \frac{2i\varepsilon}{\gamma}, \quad \langle a^\dagger \rangle_{ss} = -\frac{2i\varepsilon}{\gamma} \Rightarrow \langle a^\dagger a \rangle_{ss} = \frac{i\varepsilon}{\gamma} \left(-\frac{4i\varepsilon}{\gamma} \right) = \frac{4\varepsilon^2}{\gamma^2}$$

* Now notice that in steady state $\langle a^\dagger a \rangle = |\langle a \rangle|^2$

Notice that for coherent state $|\alpha\rangle$: $\langle \alpha | a | \alpha \rangle = \alpha$, $\langle \alpha | a^\dagger | \alpha \rangle = \alpha^*$
 $\langle \alpha | a^\dagger a | \alpha \rangle = |\alpha|^2$

which means $\langle \alpha | a^\dagger a | \alpha \rangle = |\langle \alpha | a | \alpha \rangle|^2$

Thus steady state solution is a coherent state $|\alpha\rangle$ with $\alpha = \frac{2i\varepsilon}{\gamma}$

Remark: To prove $\langle a^\dagger a \rangle = |\langle a \rangle|^2$ uniquely defines the coherent state,
 Let $\delta a = a - \langle a \rangle$ be the fluctuation operator.

$$\langle \delta a^\dagger \delta a \rangle = \langle (a^\dagger - \langle a \rangle^*)(a - \langle a \rangle) \rangle = \langle a^\dagger a \rangle - |\langle a \rangle|^2 = 0$$

$$\text{Since } \langle |\delta a|^2 \rangle = 0 \Rightarrow \delta a = 0$$

$$\text{Then } \delta a |\Psi\rangle = a |\Psi\rangle - \langle a \rangle |\Psi\rangle = 0 \Rightarrow a |\Psi\rangle = \langle a \rangle |\Psi\rangle$$

— $\int$

$\Downarrow$
 This is the eigenvalue equation of the coherent state $|\Psi\rangle = |\alpha\rangle = |\langle a \rangle\rangle$

Just show

$$f = |\alpha\rangle \langle \alpha|$$

Satisfy the Master equation.

2) Bloch equations in the interaction picture (rotating frame - $\omega_0 = 0$):

$$\left. \begin{aligned} \frac{d\langle\sigma_z\rangle}{dt} &= -\gamma(\langle\sigma_z\rangle + 1) \\ \frac{d\langle\sigma_x\rangle}{dt} &= -\left(\frac{\gamma}{2} + \gamma_p\right)\langle\sigma_x\rangle \\ \frac{d\langle\sigma_y\rangle}{dt} &= -\left(\frac{\gamma}{2} + \gamma_p\right)\langle\sigma_y\rangle \end{aligned} \right\} \begin{aligned} \sigma_{\pm} &= \frac{1}{2}(\sigma_x \pm i\sigma_y) \\ \frac{d\langle\sigma_{\pm}\rangle}{dt} &= -\left(\frac{\gamma}{2} + \gamma_p\right)\langle\sigma_{\pm}\rangle \\ \langle\sigma_{\pm}(t)\rangle &= \langle\sigma_{\pm}(0)\rangle e^{-\left(\frac{\gamma}{2} + \gamma_p\right)t} \end{aligned}$$

Thus we have $\langle\sigma_{\pm}(t+\tau)\rangle = e^{-\left(\frac{\gamma}{2} + \gamma_p\right)\tau} \langle\sigma_{\pm}(t)\rangle$

Applying quantum regression thm., we multiply both sides of expectation value by $\langle\sigma_{-}(t)\rangle$.

$$\langle\sigma_{+}(t+\tau)\sigma_{-}(t)\rangle = e^{-\left(\frac{\gamma}{2} + \gamma_p\right)\tau} \langle\sigma_{+}(t)\sigma_{-}(t)\rangle$$

$$\sigma_{+}\sigma_{-} = |e\rangle\langle g|g\rangle\langle e| = |e\rangle\langle e| \Rightarrow \sigma_{+}(t)\sigma_{-}(t) = p_e(t) = |e\rangle\langle e|$$

From eq. 12, $\frac{dp_e}{dt} = -\gamma p_e(t) \Rightarrow p_e(t) = e^{-\gamma t} p_e(0)$

Remark: Note that even after decoherence effect is introduced it won't effect excited state population.

As noted in eq. 15, $H_{col} \propto \sigma_z$. Because σ_z commutes with free Hamiltonian, these collisions are energy conserving. The random collisions destroy coherences $p_{eg} = |e\rangle\langle g|$, $p_{ge} = |g\rangle\langle e|$ but doesn't cause transitions btwn. energy levels. Another way to see this is to calculate

$$[\sigma_z, [\sigma_z, \rho]] = \left[\sigma_z, \begin{pmatrix} 0 & 2p_{eg} \\ 2p_{ge} & 0 \end{pmatrix} \right] = \begin{bmatrix} 0 & 4p_{eg} \\ 4p_{ge} & 0 \end{bmatrix} \text{ in the dephasing included dynamics. (Equation 17.)}$$

Since diagonal terms are 0, $\rho_e(t)$ will only depend on simultaneous emissions, not dephasing due to atomic collisions.

$$\text{Thus, } \langle \sigma_+(t) \sigma_-(t) \rangle = e^{-\gamma t} \langle \sigma_+(0) \sigma_-(0) \rangle$$

$$\langle \sigma_+(t+\tau) \sigma_-(t) \rangle = e^{(-\frac{\gamma}{2} - \gamma_p)\tau} e^{-\gamma t} \langle \sigma_+(0) \sigma_-(0) \rangle //$$

*Note that in laboratory frame, an extra $e^{-i\omega_0\tau}$ term would appear.

→ Now to calculate $\langle \sigma_+(t) \sigma_+(t+\tau) \sigma_-(t+\tau) \sigma_-(t) \rangle$, notice from earlier

$$\sigma_+(t+\tau) \sigma_-(t+\tau) = \rho_e(t+\tau) = e^{-\gamma\tau} \rho_e(t) = e^{-\gamma\tau} \langle \sigma_+ \sigma_- \rangle(t)$$

Now applying the quantum regression thm, allows us to sandwich this evolution between earlier time operators $\sigma_+(t)$ and $\sigma_-(t)$

$$\langle \sigma_+(t) [\sigma_+(t+\tau) \sigma_-(t+\tau)] \sigma_-(t) \rangle = e^{-\gamma\tau} \langle \sigma_+ (\sigma_+ \sigma_-) \sigma_- \rangle(t)$$

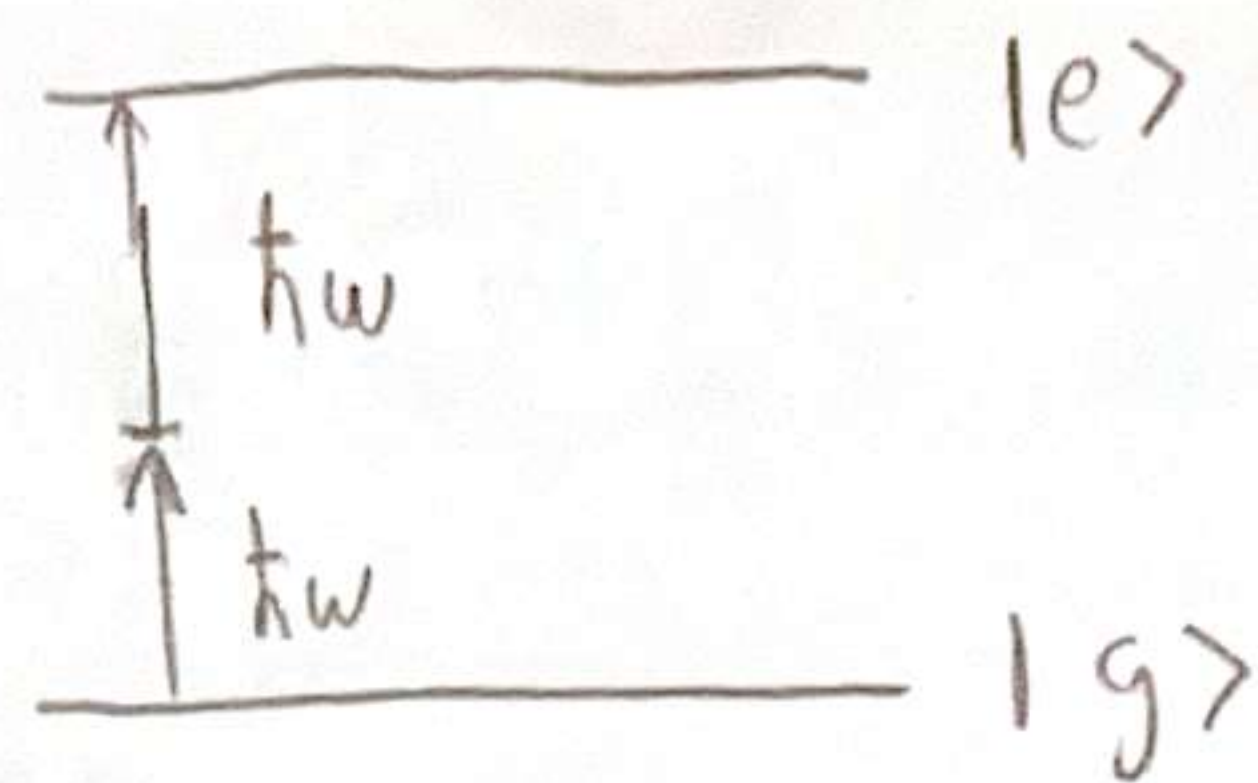
Notice that $\sigma_+^2 = |e\rangle\langle g|e\rangle\langle g| = 0$, $\sigma_-^2 = |g\rangle\langle e|g\rangle\langle e| = 0$

Therefore $\langle \sigma_+(t) \sigma_+(t+\tau) \sigma_-(t+\tau) \sigma_-(t) \rangle = 0$ ★

! This result is the consequence of photon blockade effect.

If atom emits/absorbs a photon at time t , it cannot spontaneously emit/absorb a second photon at $t+\tau$. Thus 2-photon coincidence is strictly zero.

$$3) H_{\text{eff}} = \hbar \chi (a^2 \sigma_+ + a^{\dagger 2} \sigma_-)$$



$\hookrightarrow a^2 \sigma_-$ destroys atomic excitation ($|e\rangle \rightarrow |g\rangle$) and creates 2 photons in the cavity

$\hookrightarrow a^{\dagger 2} \sigma_+$ destroy 2 photons in the cavity and creates an atomic excitation.

$\Rightarrow$ Thus Hamiltonian couples to specific pairs of states. If atom is excited and there are n photons, state is $|n, e\rangle$. This couples to state where atom is in ground state and has $n+2$ photons $\Rightarrow |n+2, g\rangle$

Thus system breaks down to 2-D subspaces spanned by:

$$\mathcal{H} = \{ |n, e\rangle, |n+2, g\rangle \} \quad , \quad n \geq 0$$

a)

$$H_{\text{eff}} |n, e\rangle = \hbar \chi a^2 \sigma_- |n, e\rangle = \hbar \chi \sqrt{n+1} \sqrt{n+2} |n+2, g\rangle$$

$$H_{\text{eff}} |n+2, g\rangle = \hbar \chi a^{\dagger 2} \sigma_+ |n+2, g\rangle = \hbar \chi \sqrt{n+2} \sqrt{n+1} |n, e\rangle$$

In the $\{ |n, e\rangle, |n+2, g\rangle \}$ basis $\Rightarrow H_n = \hbar \Omega_n \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$\text{where } \Omega_n = \chi \sqrt{(n+1)(n+2)}$$

Eigen values of this H_n matrix are simply $E_{\pm, n} = \pm \hbar \Omega_n$ with eigenvectors as dressed states:

$$|n, \pm\rangle = \frac{1}{\sqrt{2}} (|n, e\rangle \pm |n+2, g\rangle)$$

b) Initially atom is in ground state and field is in a number state $|m\rangle$:

$$|\Psi(0)\rangle = |m, g\rangle$$

To apply time evolution, let us rewrite $|m, g\rangle$ in terms of dressed states $|n, \pm\rangle = \frac{1}{\sqrt{2}} (|n, e\rangle \pm |n+2, g\rangle)$

$$|m-2, \pm\rangle = \frac{1}{\sqrt{2}} (|m-2, e\rangle \pm |m, g\rangle)$$

$$\text{Then, } |\Psi(0)\rangle = \frac{1}{\sqrt{2}} (|m-2, +\rangle - |m-2, -\rangle)$$

Since dressed states are stationary eigenstates:

$$\begin{aligned} |\Psi(t)\rangle &= \frac{1}{\sqrt{2}} \left(\underbrace{|m-2, +\rangle}_{\frac{1}{\sqrt{2}}(|m-2, e\rangle + |m, g\rangle)} e^{-i\Omega_{m-2}t} - \underbrace{|m-2, -\rangle}_{\frac{1}{\sqrt{2}}(|m-2, e\rangle - |m, g\rangle)} e^{+i\Omega_{m-2}t} \right) \\ &= \frac{1}{2} (e^{-i\Omega_{m-2}t} - e^{+i\Omega_{m-2}t}) |m-2, e\rangle + \frac{1}{2} (e^{-i\Omega_{m-2}t} + e^{+i\Omega_{m-2}t}) |m, g\rangle \\ &= -i \sin(\Omega_{m-2}t) |m-2, e\rangle + \cos(\Omega_{m-2}t) |m, g\rangle \end{aligned}$$

$$|\Psi(t)\rangle = [-i \sin(\Omega_{m-2}t), \cos(\Omega_{m-2}t)]^T \text{ in 2D Hilbert basis.}$$

$$\begin{aligned} \langle \Psi(t) | \hat{\sigma}_z | \Psi(t) \rangle &= [i \sin(\Omega_{m-2}t), \cos(\Omega_{m-2}t)] \begin{bmatrix} -i \sin(\Omega_{m-2}t) \\ -\cos(\Omega_{m-2}t) \end{bmatrix} \\ &= \sin^2(\Omega_{m-2}t) - \cos^2(\Omega_{m-2}t) = -\cos(2\Omega_{m-2}t) \end{aligned}$$

Recalling $\Omega_m = \hbar \sqrt{(m-1)m} \Rightarrow$ Rabi frequency

$$\langle \Psi(t) | \hat{\sigma}_z | \Psi(t) \rangle = \boxed{-\cos(2\hbar t \sqrt{m(m-1)})}$$

4) First we define Rabi frequencies, that describe interaction:

a) $\Omega_{pa} = 2g_{pa} \alpha_p \Rightarrow$ interaction btwn. p field and $|a\rangle$ state
 $\Omega_{pa'} = 2g_{pa'} \alpha_p \Rightarrow$ p $|a'\rangle$
 $\Omega_{la} = 2g_{la} \alpha_l \Rightarrow$ l $|a\rangle$
 $\Omega_{la'} = 2g_{la'} \alpha_l \Rightarrow$ l $|a'\rangle$

$\hookrightarrow$ 2 factor comes from $\cos \Omega t = \frac{1}{2}(e^{i\Omega t} + e^{-i\Omega t})$

Unperturbed, free energy Hamiltonian is

$$H_0 = \hbar \omega_a |a\rangle \langle a| + \hbar \omega_{a'} |a'\rangle \langle a'| + \hbar \omega_c |c\rangle \langle c|$$

where we assume $\omega_b = 0$. Energy of ground state $|b\rangle$ can be assumed 0 (or do a RWA to realize).

Interaction Hamiltonian is:

$$H_I = \frac{\hbar}{2} \left\{ \Omega_{pa} e^{-i\omega_p t} |a\rangle \langle b| + \Omega_{pa'} e^{-i\omega_p t} |a'\rangle \langle b| \right. \\ \left. + \Omega_{la} e^{-i\omega_l t} |a\rangle \langle c| + \Omega_{la'} e^{-i\omega_l t} |a'\rangle \langle c| \right\} + \text{hermitian conjugate (h.c.)}$$

$\Rightarrow$ transitions from $|b\rangle$ to $|a\rangle, |a'\rangle$
 $\Rightarrow$ transitions from $|c\rangle$ to $|a\rangle, |a'\rangle$

Let us make the following rotating frame transformations

$$|a\rangle \rightarrow e^{-i\omega_p t} |a\rangle$$

$$|a'\rangle \rightarrow e^{-i\omega_p t} |a'\rangle$$

$$|c\rangle \rightarrow e^{-i(\omega_p - \omega_l)t} |c\rangle$$

First interaction Hamiltonian transform as:

$$H_I = \frac{\hbar}{2} \left(\Omega_{pa} e^{-i\omega_p t} e^{i\omega_p t} |a\rangle \langle b| + \Omega_{pa'} e^{-i\omega_p t} e^{i\omega_p t} |a'\rangle \langle b| \right. \\ \left. + \Omega_{la} e^{-i\omega_l t} e^{i\omega_p t} e^{-i(\omega_p - \omega_l)t} |a\rangle \langle c| \right. \\ \left. + \Omega_{la'} e^{-i\omega_l t} e^{i\omega_p t} e^{-i(\omega_p - \omega_l)t} |a'\rangle \langle c| \right) + \text{h.c.}$$

$$H_I = \frac{\hbar}{2} \left(\Omega_{pa} |a\rangle \langle b| + \Omega_{pa'} |a'\rangle \langle b| + \Omega_{la} |a\rangle \langle c| + \Omega_{la'} |a'\rangle \langle c| \right)$$

Now in the free energy term we correspondingly shift $w_{a(a')}$ by w_p , and w_c by $w_p - w_l$.

- $w_a \rightarrow \Delta_a = w_a - w_p$
- $w_{a'} \rightarrow \Delta_{a'} = w_{a'} - w_p$

• $w_c \rightarrow w_c - (w_p - w_l)$

↳ Notice $w_{ab} - w_p = w_{ac} - w_l$ ($w_{ab} = w_a$)

$w_a - w_p = w_a - w_c - w_l \Rightarrow w_c = w_p - w_l$

↳ Thus $w_c \rightarrow 0$

* Note that we could've done this rotating frame transformation more formally by:

$$H_{\text{eff}} = U^\dagger H U - i\hbar U^\dagger \dot{U} \quad \text{where} \quad U(t) = \exp \left[-i \left(w_p |a\rangle \langle a| + w_p |a'\rangle \langle a'| + (w_p - w_l) |c\rangle \langle c| \right) \right]$$

Total effective Hamiltonian is then:

$$H_{\text{eff}} = \hbar \Delta_a |a\rangle \langle a| + \hbar \Delta_{a'} |a'\rangle \langle a'| + \frac{\hbar}{2} \left(\Omega_{pa} |a\rangle \langle b| + \Omega_{pa'} |a'\rangle \langle b| + \Omega_{la} |a\rangle \langle c| + \Omega_{la'} |a'\rangle \langle c| \right) + \text{h.c.}$$

* Remark 1: $\frac{\Omega_{ij}}{2} = g_{ij} \propto_i$ where $i \in \{p, l\}$, $j \in \{a, a'\}$

* Remark 2: $\Delta_c = w_c - (w_p - w_l) = 0$ shows that $|c\rangle$ is also a zero energy state degenerate with $|b\rangle$.

b) Dark state $|\Psi_0\rangle$ is an eigenstate of H_{eff} with eigenval 0

$$H|\Psi_0\rangle = 0$$

Start by assuming $|\Psi_0\rangle = C_a|a\rangle + C_{a'}|a'\rangle + C_b|b\rangle + C_c|c\rangle$

$$\begin{aligned} H_{\text{eff}}|\Psi_0\rangle = & (C_a \hbar \Delta_a + C_b \hbar \frac{\Omega_{pa}}{2} + C_c \hbar \frac{\Omega_{la}}{2}) |a\rangle \\ & (C_{a'} \hbar \Delta_{a'} + C_b \hbar \frac{\Omega_{pa'}}{2} + C_c \hbar \frac{\Omega_{la'}}{2}) |a'\rangle \\ & + (C_a \hbar \frac{\Omega_{pa}}{2} + C_{a'} \hbar \frac{\Omega_{la}}{2}) |b\rangle \\ & + (C_{a'} \hbar \frac{\Omega_{pa'}}{2} + C_a \hbar \frac{\Omega_{la'}}{2}) |c\rangle = 0 \end{aligned}$$

→ For coeffs. of $|b\rangle$ and $|c\rangle$ terms to be simultaneously zero we require $C_a = C_{a'} = 0$

↳ Physically: there should not be any any decaying states $|a\rangle, |a'\rangle$ in the superposition of dark states

Then for coeffs. of $|a\rangle$ and $|b\rangle$ to be zero, we need

$$\frac{\Omega_{pa}}{\Omega_{pa'}} = \frac{\Omega_{la}}{\Omega_{la'}} \Rightarrow \text{existence condition:}$$

$$\Downarrow$$

$$C_b \Omega_{pa} = -C_c \Omega_{la}$$

If transitions rate from $|b\rangle$ to $|a\rangle$ followed by $|a'\rangle$ to $|c\rangle$ is equal to rate from $|c\rangle$ to $|a\rangle$ followed by $|a'\rangle$ to $|b\rangle$, then effectively it's as if photons emitted from transitions are all absorbed again by the system. Hence the total system appears "dark".

$$|\Psi_0\rangle = \frac{-\Omega_{la}|b\rangle - \Omega_{pa}|c\rangle}{\sqrt{|\Omega_{la}|^2 + |\Omega_{pa}|^2}}$$

$$= \frac{g_{la} \propto_l |b\rangle - g_{pa} \propto_p |c\rangle}{\sqrt{(g_{la} \propto_l)^2 + (g_{pa} \propto_p)^2}}$$

$$5) \hat{L}_{\text{grav}} = \sqrt{\Lambda_{\text{grav}}} (b+b^\dagger) \quad , \quad \Lambda_{\text{grav}} = \frac{2\pi}{3} \frac{G\Delta}{W_m}$$

$$\frac{d\rho}{dt} = -\frac{i}{\hbar} [H_0, \rho] + 2 L_{\text{grav}} \rho L_{\text{grav}}^\dagger - \{ L_{\text{grav}}^\dagger L_{\text{grav}} \rho \}$$

For harmonic oscillator, $H_0 = \hbar W_m b^\dagger b$ and $L_{\text{grav}} = L_{\text{grav}}^\dagger$ Hermitian.

$$\frac{d\rho}{dt} = -iW_m [b^\dagger b, \rho] + \Lambda_{\text{grav}} [2(b+b^\dagger)\rho(b+b^\dagger) - (b+b^\dagger)^2 \rho - \rho(b+b^\dagger)^2]$$

$$\frac{d\langle b^\dagger b(t) \rangle}{dt} = \text{Tr} \left(b^\dagger b \frac{d\rho}{dt} \right) = -iW_m \text{tr} \left((b^\dagger b)^2 \rho - b^\dagger b \rho b^\dagger b \right)$$

$$+ \Lambda_{\text{grav}} [2\text{tr}(b^\dagger b (b+b^\dagger)\rho(b+b^\dagger)) - b^\dagger b (b+b^\dagger)^2 \rho - b^\dagger b \rho (b+b^\dagger)^2]$$

$$\hookrightarrow -iW_m \text{tr} \left((b^\dagger b)^2 \rho - (b^\dagger b)^2 \rho \right) = 0 \Rightarrow \text{first term}$$

$$\hookrightarrow \text{tr} \left((2(b+b^\dagger) b^\dagger b (b+b^\dagger) - b^\dagger b (b+b^\dagger)^2 - (b+b^\dagger)^2 b^\dagger b) \rho \right)$$

$$= \text{tr} \left(([b+b^\dagger, b^\dagger b] (b+b^\dagger) + (b+b^\dagger)^\dagger [b b^\dagger, b+b^\dagger]) \rho \right)$$

$$\times [b+b^\dagger, b^\dagger b] = (b+b^\dagger) b^\dagger b - b^\dagger b (b+b^\dagger) = b b^\dagger b + b^{\dagger 2} b - b^\dagger b^2 - b^\dagger b b^\dagger$$

$$= [b, b^\dagger] b + b^\dagger [b^\dagger, b] = b - b^\dagger$$

$$= \text{tr} \left((b - b^\dagger)(b+b^\dagger) - (b+b^\dagger)(b - b^\dagger) \right) = b^2 - b^{\dagger 2} + [b, b^\dagger]$$

$$- (b^2 - b^{\dagger 2} - [b, b^\dagger]) = 2$$

$$\text{Thus } \frac{d\langle b^\dagger b(t) \rangle}{dt} = \frac{d\langle n(t) \rangle}{dt} = 2\Lambda_{\text{grav}}$$

$$\boxed{\langle n(t) \rangle = \langle n(0) \rangle + 2\Lambda_{\text{grav}} t} \Rightarrow$$

Since $\Lambda_{\text{grav}} > 0$, mean phonon number grows linearly in time. This injection of energy demonstrates heating effect.

b) In lecture 14, (equation 10), we've derived how system evolves when intrinsic decay is present:

$$\frac{d\langle n \rangle}{dt} = -\gamma \langle n \rangle + \gamma_m N$$

Since our aim is to isolate phonon number caused by gravitational coherence we can assume that system is in perfect vacuum at absolute zero and hence $N=0$. Since dissipation terms don't interact with each other, they are additive

$$\frac{d\langle n \rangle}{dt} = -\gamma \langle n \rangle + 2\Delta_{\text{grav}} \Rightarrow \langle n \rangle_{ss} = \frac{2\Delta_{\text{grav}}}{\gamma_m}$$

$$\Delta_{\text{grav}} = \frac{2\pi}{3} \frac{G\Delta}{\omega_m} = \frac{(6.674 \times 10^{-11}) (2500)}{3 \cdot 1} \approx 5.56 \times 10^{-8} \text{ s}^{-1}$$

$$\gamma_m = \frac{\omega_m}{Q} = \frac{2\pi}{10^{10}} \approx 6.28 \times 10^{-10} \text{ s}^{-1}$$

$$\langle n \rangle_{ss} = \frac{2 \times (5.56 \times 10^{-8})}{6.28 \times 10^{-10}} = \boxed{177.1 \text{ phonons}}$$

* To observe gravitational heating effect, phonons produced by grav. decoherence must be at least equal to # of thermal phonons.

$$n_{\text{thermal}} = \frac{k_B T}{\hbar \omega_m} = 177.1 \Rightarrow T \approx \frac{177.1 \times (1.05 \times 10^{-34} \text{ J.s}) \times (2\pi \text{ rad/s})}{1.38 \times 10^{-23} \text{ J/K}}$$

$$T \approx 8.5 \times 10^{-9} \text{ K} = \underline{\underline{8.5 \text{ nK}}}$$

c) Writing the Lindblad operator in $\hat{x}$ basis

$$\hat{x} = x_{\text{ZPF}} (b + b^\dagger) \Rightarrow L_{\text{grav}} = \frac{\sqrt{\Lambda_{\text{grav}}}}{x_{\text{ZPF}}} \hat{x}$$

$$\frac{dp}{dt} = -\frac{i}{\hbar} [H_0, p] + \frac{\Lambda_{\text{grav}}}{x_{\text{ZPF}}^2} (2\hat{x} p \hat{x} - \hat{x}^2 p - p \hat{x}^2)$$

To find the decoherence of $|x\rangle, |x'\rangle$ states, we are interested in

$$\begin{aligned} \frac{d\rho_{xx'}}{dt} &= \langle x' | \frac{dp}{dt} | x \rangle = -\frac{i}{\hbar} \langle x' | [H_0, p] | x \rangle \\ &\quad + \frac{\Lambda_{\text{grav}}}{x_{\text{ZPF}}^2} (2x'x \langle x' | p | x \rangle - x'^2 \langle x' | p | x \rangle - x^2 \langle x' | p | x \rangle) \end{aligned}$$

$$\frac{d}{dt} \langle x' | p | x \rangle = -\frac{i}{\hbar} \underbrace{\langle x' | [H_0, p] | x \rangle}_{\text{trace will give zero or const}} - \underbrace{\frac{\Lambda_{\text{grav}}}{x_{\text{ZPF}}^2} (x-x')^2}_{\text{decoherence term}} \langle x' | p | x \rangle$$

$$T_{\text{dec}} = \frac{\Lambda_{\text{grav}}}{x_{\text{ZPF}}^2} |x-x'|^2 \quad ; \text{ decoherence rate}$$

$$\hookrightarrow \text{Case 1: } |x-x'| = x_{\text{ZPF}} \Rightarrow T_{\text{dec}} = \Lambda_{\text{grav}} = 5.56 \times 10^{-8} \text{ s}^{-1}$$

$\hookrightarrow$ extremely slow, correlation survive for a long time

$$\hookrightarrow \text{Case 2: } |x-x'| = 1 \Rightarrow T_{\text{dec}} = \frac{2m\omega_m \Lambda_{\text{grav}}}{\hbar} = \frac{5.56 \times 10^{-8}}{2.10 \times 10^{-37}} \approx 2.64 \times 10^{29} \text{ s}^{-1}$$

$\hookrightarrow$ Very fast decay rate. This demonstrates that gravity destroys macroscopic superpositions instantaneously. (almost)